

# An Empirical Framework for the Health-Cost Incidence, Persistence, and Distributional Burden of Climate Shocks

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June 2026

## 1 Setting and identification

The analysis uses two panels that share a common county-level climate-shock series but measure different outcome domains. *Panel A* is a county  $\times$  year panel (demand side):  $\sim 41,400$  county-years, 3,155 counties, 2011–2023, with outcomes for individual-market benchmark premiums, medical-debt share, real personal income per capita, and civilian employment. *Panel B* is a hospital  $\times$  year panel (supply side):  $\sim 59,900$  hospital-years, 5,119 hospitals (CMS Certification Numbers), with operating and net margins and uncompensated care. Each hospital is geo-coded to a single county (modal county per ZIP, facility location) and inherits that county’s shocks, so  $S_{it} \equiv S_{c(i),t}$ . Both panels carry county (or hospital) and year fixed effects and cluster standard errors at the state level ( $\sim 50$  clusters).

All papers use four binary shocks defined at the county-year level: `ls_Extreme_Drought` (Palmer PDSI  $\leq -4$ ); `High_CDD / High_HDD` (cooling/heating degree-days in the top quartile of the county’s own historical distribution); and `High_AQI_Max` (annual maximum AQI  $> 100$ ). Generically  $S_{ct} \in \{0, 1\}$ . These are *recurring* shocks, not one-time absorbing treatments: a county may enter, leave, and re-enter drought. This rules out canonical staggered-adoption DiD as the *primary* design and motivates the persistence estimators of Section 3; Section 2 adds a first-onset cohort DiD as a contamination-free cross-check. Defining shocks against each county’s own 1990–2000 baseline distribution makes “shock” an unusual draw relative to local historical norms rather than an absolute threshold.

## 2 Paper 1 — Incidence: do climate shocks raise health costs?

### 2.1 Distributed-lag two-way fixed effects

Paper 1 asks whether contemporaneous and lagged exposure shifts the *level* of the outcome. For outcome  $Y$  and shock  $S$ ,

$$Y_{ct} = \alpha_c + \gamma_t + \sum_{k=0}^2 \beta_k S_{c,t-k} + \mathbf{X}'_{ct} \delta + \varepsilon_{ct}, \quad (1)$$

with county FE  $\alpha_c$ , year FE  $\gamma_t$ , and time-varying controls  $\mathbf{X}_{ct}$  (log real income per capita, unemployment, Medicaid-expansion status); the hospital version replaces  $c$  with hospital  $i$ . Equation (1) is estimated *once per (outcome, shock) pair* on the full panel; the three lag terms enter the same regression. The subscripts run  $t, t-1, t-2$  because a shock can only affect current and *subsequent* outcomes: a single drought in year  $t$  appears as the  $\beta_0$  regressor for the year- $t$  outcome, the  $\beta_1$  regressor for the year- $(t+1)$  outcome, and the  $\beta_2$  regressor for year  $t+2$ . The identifying variation is counties that *switch* shock status; always- and never-shocked counties are absorbed by  $\alpha_c$ . The cumulative (impulse-response) effect of one shock-year traced across all three horizons is

$$\hat{\beta}_{\text{cum}} = \hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2, \quad \widehat{\text{Var}}(\hat{\beta}_{\text{cum}}) = \sum_{j=0}^2 \sum_{k=0}^2 \widehat{\text{Cov}}(\hat{\beta}_j, \hat{\beta}_k), \quad (2)$$

the off-diagonal covariances taken from the state-clustered vcov via a Wald linear-combination test. On the hospital panel, heat raises uncompensated care (High\_CDD: +1.22 pp of net patient revenue,  $p = 0.063$ ; cold: +\$1.55M per hospital,  $p = 0.022$ ). The negative drought coefficient on uncompensated care ( $-1.68$  pp,  $p = 0.032$ ) is a measurement artifact—billing and credit-bureau measures undercount distress in drought-stricken, lower-utilization areas—and is flagged throughout rather than read causally.

## 2.2 A sharper control group: the 2012 drought natural experiment

Because the lag model’s control group for any drought year silently includes counties droughted in *other* years, we add a first-onset cohort DiD. Let  $\text{Treated}_c = 1$  if county  $c$ ’s first-ever extreme-drought year in the panel is 2012, and  $\text{Post}_t = \mathbf{1}[t \geq 2012]$ :

$$Y_{ct} = \alpha_c + \gamma_t + \tau (\text{Treated}_c \times \text{Post}_t) + \varepsilon_{ct}. \quad (3)$$

The FEs absorb both main effects, so  $\hat{\tau}$  is the ATT: 139 first-onset-2012 counties against 2,534 *never*-droughted controls (counties first droughted in other years are excluded, which is what cleans the control group). With a single pre-period (2011) the regression returns the  $2 \times 2$  difference exactly.

The sharp  $2 \times 2$  assumes *unconditional* parallel trends, which is strong here: treated counties skew rural-agricultural (Georgia, the Mountain West, the Plains) while controls skew urban and coastal. We therefore also report the Sant’ Anna–Zhao (2020) doubly-robust ATT, consistent if *either* the propensity or the outcome-regression model is correct, conditioning only on *baseline* (2011) covariates—log population, baseline median household income, Census division—and deliberately not on contemporaneous income, employment, or premiums, which are post-treatment mediators (bad controls):

$$\hat{\tau}_{\text{dr}} = \mathbb{E} \left[ \left( \frac{D}{\mathbb{E}[D]} - \frac{\hat{\pi}(X)(1-D)/(1-\hat{\pi}(X))}{\mathbb{E}[\hat{\pi}(X)(1-D)/(1-\hat{\pi}(X))]} \right) (\Delta Y - \hat{m}_0(X)) \right] \quad (4)$$

where  $\Delta Y$  is the pre-to-post outcome change (post collapsed to the 2012–2023 mean),  $\hat{\pi}(X)$  the propensity score, and  $\hat{m}_0(X)$  the control outcome-change regression. Results are in Table 1.

**Table 1:** Incidence: 2012-drought natural experiment (county panel, \$2023)

| Outcome                     | Uncond. $2 \times 2$ $\hat{\tau}$ | Doubly-robust $\hat{\tau}_{\text{dr}}$ | DR 95% CI        |
|-----------------------------|-----------------------------------|----------------------------------------|------------------|
| Personal income per capita  | −1,311                            | − <b>1,451</b>                         | [−2,461, −441]   |
| Civilian employed (persons) | −2,053                            | −871                                   | [−1,719, −23]    |
| Median household income     | −992                              | −1,186                                 | [−2,140, −232]   |
| Medical-debt share          | −0.006 (n.s.)                     | −0.011                                 | [−0.019, −0.002] |

139 treated (first-onset 2012) vs. 2,534 never-exposed; state-clustered.  $2 \times 2$ : income  $z = -2.28$  ( $p = 0.027$ ), employment  $z = -4.29$  ( $p = 0.0001$ ). Income is robust to conditioning (if anything stronger); employment attenuates  $\sim 58\%$  and is the composition-sensitive secondary finding. Medical debt is measurement-fragile. Re-estimated as a full Callaway–Sant’ Anna model across *all* drought cohorts the average is null (PCPI +\$350, s.e. 585), so the negative real-economy effect is specific to the large, clean 2012 event and is an intent-to-treat “effect of first onset.”

## 3 Paper 2 — Persistence: scarring, reversibility, dose

### 3.1 Onset/persist/exit local projections

Converting the binary series into a transition state from  $(S_{ct}, S_{c,t-1})$  gives *onset*  $D^{on} = (1, 0)$ , *persist*  $D^{per} = (1, 1)$ , and *exit*  $D^{ex} = (0, 1)$ , with  $(0, 0)$  omitted as the reference. Following Jordà (2005), a separate regression is run per horizon  $h$ :

$$Y_{i,t+h} = \alpha_i + \gamma_t + \beta_h^{on} D_{it}^{on} + \beta_h^{per} D_{it}^{per} + \beta_h^{ex} D_{it}^{ex} + \varepsilon_{i,t+h}, \quad h = 0, 1, 2, \quad (5)$$

where the LHS is the outcome  $h$  years after the transition year  $t$  and the dummies are always indexed by  $t$ . Each coefficient is measured against the never-shocked  $(0,0)$  reference:  $\beta_h^{on}$  is the entry effect,  $\beta_h^{per}$  the interior of a sustained spell, and  $\beta_h^{ex}$  the first post-shock year. If a shock were fully reversible the entry cost would be exactly undone on exit, so the null and the asymmetry statistic are

$$H_0 : \beta_h^{on} + \beta_h^{ex} = 0, \quad \hat{L}_h = \hat{\beta}_h^{on} + \hat{\beta}_h^{ex}, \quad z_h = \frac{\hat{L}_h}{\sqrt{V_{11} + V_{33} + 2V_{13}}} \xrightarrow{d} \mathcal{N}(0,1), \quad (6)$$

the covariance  $V_{13}$  taken directly from the single fitted model's vcov. A nonzero  $\hat{L}_h$  means exit does not return the unit to its pre-shock trajectory—scarring.

### 3.2 Cumulative dose

A second lens asks whether the *stock* of past exposure matters. Define a never-resetting counter  $C_{it} = \sum_{s \leq t} S_{is}$  and estimate

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 C_{it} + \beta_2 C_{it}^2 + \varepsilon_{it}, \quad \left. \frac{\partial Y}{\partial C} \right|_{C=x} = \beta_1 + 2\beta_2 x, \quad (7)$$

with a binned variant  $\sum_b \delta_b \mathbf{1}[C_{it} \in b]$  over  $b \in \{1-3, 4-6, 7-9, 10+\}$  (reference  $C = 0$ ) and a high-vs-low contrast  $\hat{\Delta} = \hat{\delta}_{10+} - \hat{\delta}_{1-3}$ . Table 2 reports drought  $\rightarrow$  hospital operating margin.

**Table 2:** Persistence: extreme drought  $\rightarrow$  hospital operating margin ( $N = 57,302$ )

| Quantity                                                        | Estimate | Test                                |
|-----------------------------------------------------------------|----------|-------------------------------------|
| $\hat{\beta}_0^{on}$ (onset)                                    | +0.0159  | margin +1.59 pp vs. $(0,0)$         |
| $\hat{\beta}_0^{per}$ (persist)                                 | -0.0017  | —                                   |
| $\hat{\beta}_0^{ex}$ (exit)                                     | +0.0109  | still +1.09 pp after clearing       |
| Asymmetry $\hat{L}_0 = \hat{\beta}_0^{on} + \hat{\beta}_0^{ex}$ | +0.0268  | SE 0.0098, $z = 2.73$ , $p = 0.006$ |
| Dose marginal at $x = 5$ (Eq. 7)                                | +0.0023  | $p = 0.010$                         |
| Dose marginal at $x = 10$                                       | +0.0018  | $p = 0.14$                          |
| Binned $\hat{\Delta} = \hat{\delta}_{10+} - \hat{\delta}_{1-3}$ | +0.018   | $p = 0.033$                         |

Onset and exit are *both* positive: the margin does not unwind on exit ( $H_0$  rejected at 1%)—asymmetric persistence. The positive dose-response is read as survivorship/adaptation (financially weaker hospitals exit the unbalanced panel first), not as climate exposure improving finances; the persistence claim is anchored to the drought hysteresis result, which needs no long accumulation window.

## 4 Paper 3 — Inequality and provider heterogeneity

Paper 3 asks whether the effects of Papers 1–2 concentrate in already-disadvantaged groups. For a moderator  $M_{it}$ ,

$$Y_{it} = \alpha_i + \gamma_t + \beta S_{ct} + \theta(S_{ct} \times M_{it}) + \phi M_{it} + \mathbf{X}_{it}' \delta + \varepsilon_{it}, \quad (8)$$

so  $\beta$  is the shock effect at  $M = 0$ ,  $\beta + \theta$  at  $M = 1$  (SE from  $\sqrt{V_{\beta\beta} + V_{\theta\theta} + 2V_{\beta\theta}}$ ), and  $\theta$  the differential. The demand side uses county social vulnerability (CDC/ATSDR SVI, top-quartile dummy); the supply side uses hospital safety-net status, ownership, state Medicaid-expansion, and county hospital-market concentration (HHI). For strain-increasing outcomes (uncompensated-care %)  $\theta > 0$  indicates concentration in high- $M$  units; for operating margin a negative  $\theta$  does. Table 3 gives selected interactions.

## 5 Inference

All standard errors cluster at the state level throughout; for HIX-premium outcomes the county panel additionally reports rating-area-clustered SEs, since counties sharing a rating area have identical premiums by construction. Every post-estimation quantity—cumulative lag effects (2), asymmetry statistics (6),

**Table 3:** Heterogeneity: selected shock  $\times$  moderator interactions

| Shock $\times$ moderator            | Outcome           | Direction of $\hat{\theta}$ | $p$       |
|-------------------------------------|-------------------|-----------------------------|-----------|
| High_CDD $\times$ SafetyNet         | Uncomp. care %NPR | higher for safety-net       | 0.021     |
| Drought $\times$ SVI (county)       | Income per capita | larger loss in high-SVI     | $< 0.05$  |
| High_HDD $\times$ SafetyNet         | Uncomp. care %NPR | higher for non-safety-net   | 0.011     |
| Drought $\times$ Medicaid expansion | Uncomp. care %NPR | higher in expansion states  | $< 0.001$ |

Headline: heat-driven uncompensated care concentrates in safety-net hospitals—the supply-side analogue of high-SVI county amplification on the demand side. Interactions are shock- and outcome-specific and not all point the “vulnerability amplifies” way; all are reported transparently.

dose-level marginal effects (7), and moderator marginal effects (8)—is a linear combination  $\hat{L} = \mathbf{w}'\hat{\beta}$  with  $\widehat{\text{Var}}(\hat{L}) = \mathbf{w}'\hat{V}\mathbf{w}$  evaluated on the state-clustered vcov  $\hat{V}$ ; tests use the normal approximation and are two-tailed at  $\alpha = 0.05$ . All models are estimated with `fixest::feols` (Bergé 2018), which performs the two-way within transform and exports the clustered vcov for the `lincom` steps.

## Appendix: estimating-equation reference

| Paper                      | Estimating equation                                                                                           | Key quantity                                          |
|----------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 1 Incidence                | $Y_{it} = \alpha_i + \gamma_t + \sum_{k=0}^2 \beta_k S_{i,t-k} + \varepsilon$                                 | $\hat{\beta}_{\text{cum}} = \sum_k \hat{\beta}_k$     |
| 1 Incidence (DiD)          | $Y_{ct} = \alpha_c + \gamma_t + \tau(\text{Tr}_c \times \text{Post}_t) + \varepsilon$                         | ATT $\hat{\tau} = \Delta^T - \Delta^C$                |
| 2 Persistence (onset/exit) | $Y_{i,t+h} = \alpha_i + \gamma_t + \beta^{on} D^{on} + \beta^{per} D^{per} + \beta^{ex} D^{ex} + \varepsilon$ | $\hat{L}_h = \hat{\beta}_h^{on} + \hat{\beta}_h^{ex}$ |
| 2 Persistence (dose)       | $Y_{it} = \alpha_i + \gamma_t + \beta_1 C_{it} + \beta_2 C_{it}^2 + \varepsilon$                              | $\hat{\beta}_1 + 2\hat{\beta}_2 x$                    |
| 3 Heterogeneity            | $Y_{it} = \alpha_i + \gamma_t + \beta S_{it} + \theta(S_{it} \times M_{it}) + \phi M_{it} + \varepsilon$      | differential $\hat{\theta}$                           |

Estimation in R (`fixest`). Shock construction: `process_county_climate.R`, `process_hospital_panel.R`. Persistence helpers: `transition_symmetry.R`, `cumulative_dose.R`. Natural-experiment DiD: `run_did_analysis.R`; doubly-robust check: `did_robustness/02_doubly_robust_did.R`.